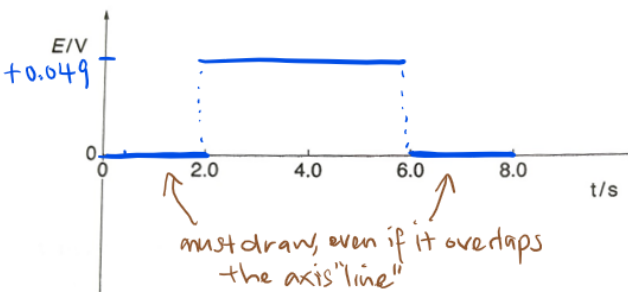


8 (a)(i)	1 mark for each of 4 pts: 1. Correct directions of uniform E & uniform B fields for a charge particle of a stated sign 2. Correct directions of electric force F_E & magnetic force F_B (they are in opposite directions) 3. Stated $F_E = F_B$ (mag) (to be undeflected & hence exit slit S_3 & be selected) 4. ie $B q v = q E$ { essential } $\textcircled{R} v = \frac{E}{B} \text{ { essential } }$	[4]
(b)(i)	• As coil rotates, there is a change in flux (linkage) • due to a change in the area of the coil perpendicular to B field By Faraday's law, an emf is induced (when there is a change in flux linking the coil)	[1] [1]
(ii)	{ For a coil rotating in a B field, instantaneous $E = NBA \omega \cos \omega t$. Hence max emf = $NBA \omega$ } Any 2 of 4: number of turns, (magnetic) flux density, area of coil & angular velocity/speed of rotation	[2]
(iii)	1. Flux varies sinusoidally with time or, $\phi = BA \cos \theta$ where θ = angle between B and normal to plane of the coil. 2. By Faraday's law/since induced emf E is equal/proportional to the rate of change of flux linkage, $\text{ie Emf} = \frac{d(N\phi)}{dt}$ $= NBA \omega \sin \theta$	[1] [1]
(c)(i)	$E = \frac{d(N\phi)}{dt} = N [B_i - B_f] A \div \Delta t$ $= 15 [0.018 - 0] (30 \times 10^{-2}) \times (24 \times 10^{-2}) \div \{ 6.0 \text{ s} - 2.0 \text{ s} \}$ $= 4.86 \times 10^{-3} = 4.9 \times 10^{-2} \text{ V}$	[1] [1]

(ii)	- From 0- 2.0 s, 6.0 – 8.0 s: horizontal line at zero V - From 2.0 – 6.0 s: horizontal line at +0.049 V, value of "0.049 V" (or 0.50 V) indicated on E- axis (ecf) } 	[1] [1]
(d)(i)	Emf induced across axle, $V = (B \sin \theta) Lv$ { straight rod } $= (6.0 \times 10^{-5}) \sin 65^\circ \times 1.5 \times 30.0$ $= 2.45 \times 10^{-3} \text{ V}$ Rate = V^2 / R $= (2.45 \times 10^{-3})^2 / 0.400 \text{ { ecf for } V }$ $= 1.50 \times 10^{-5} \text{ W}$	[1] [1] [1] [1]
(ii)	By Fleming's Left Hand Rule, the electrons in the axle are deflected by a force towards D, making D the end with a negative potential. Hence C is at a higher potential, or, By Lenz's law, the (induced) current in CD must flow in such a direction as to oppose the (rightward) motion of the rod; (hence with LEFT Thumb (induced force) pointing to left), by Fleming's LHR, the (induced) current will flow from D to C. Since rod is an emf source, C is at a higher potential.	[1] [1] [1] [1]

9(a)(i)	discrete/ packet / quantum of energy	[1]
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